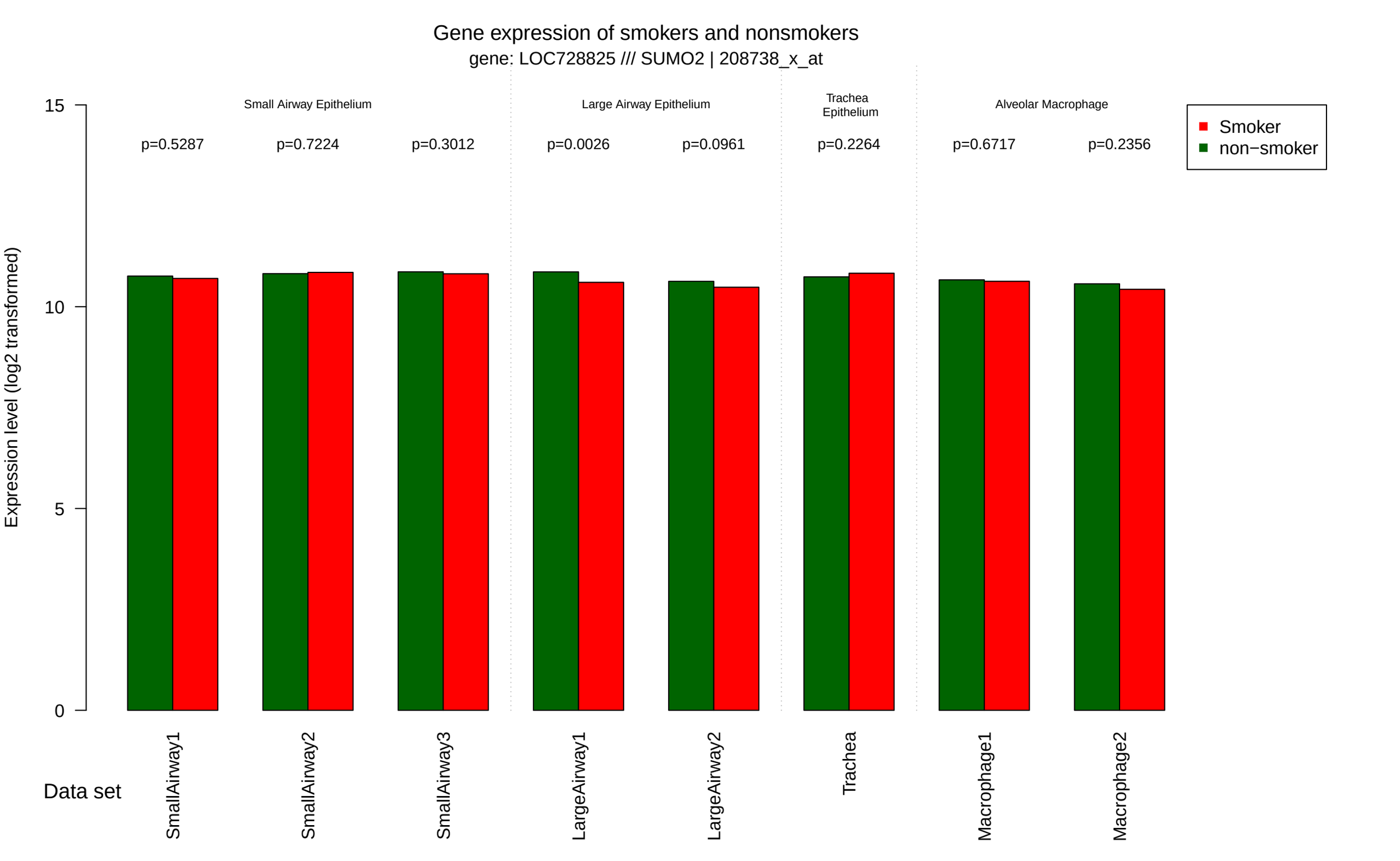
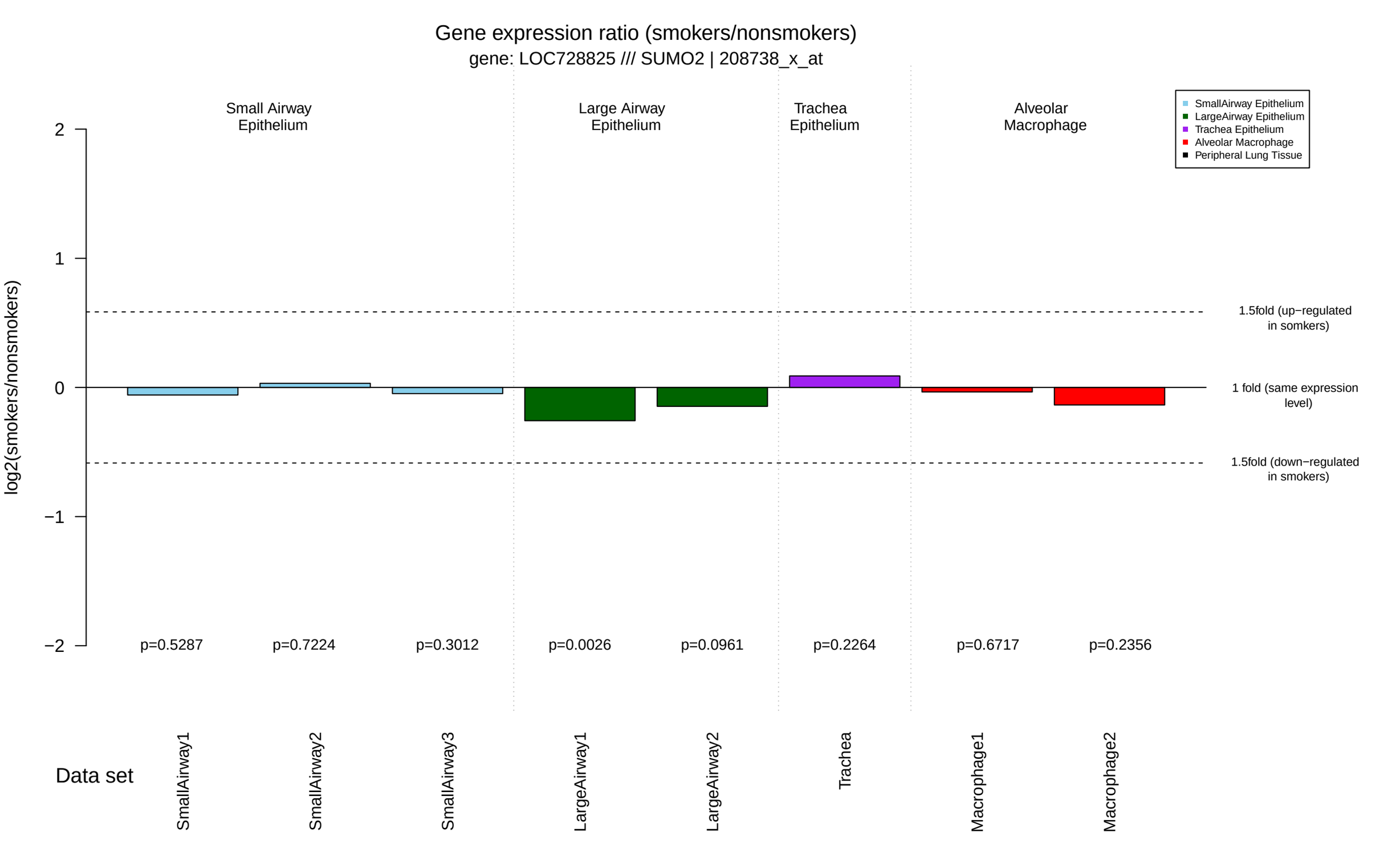
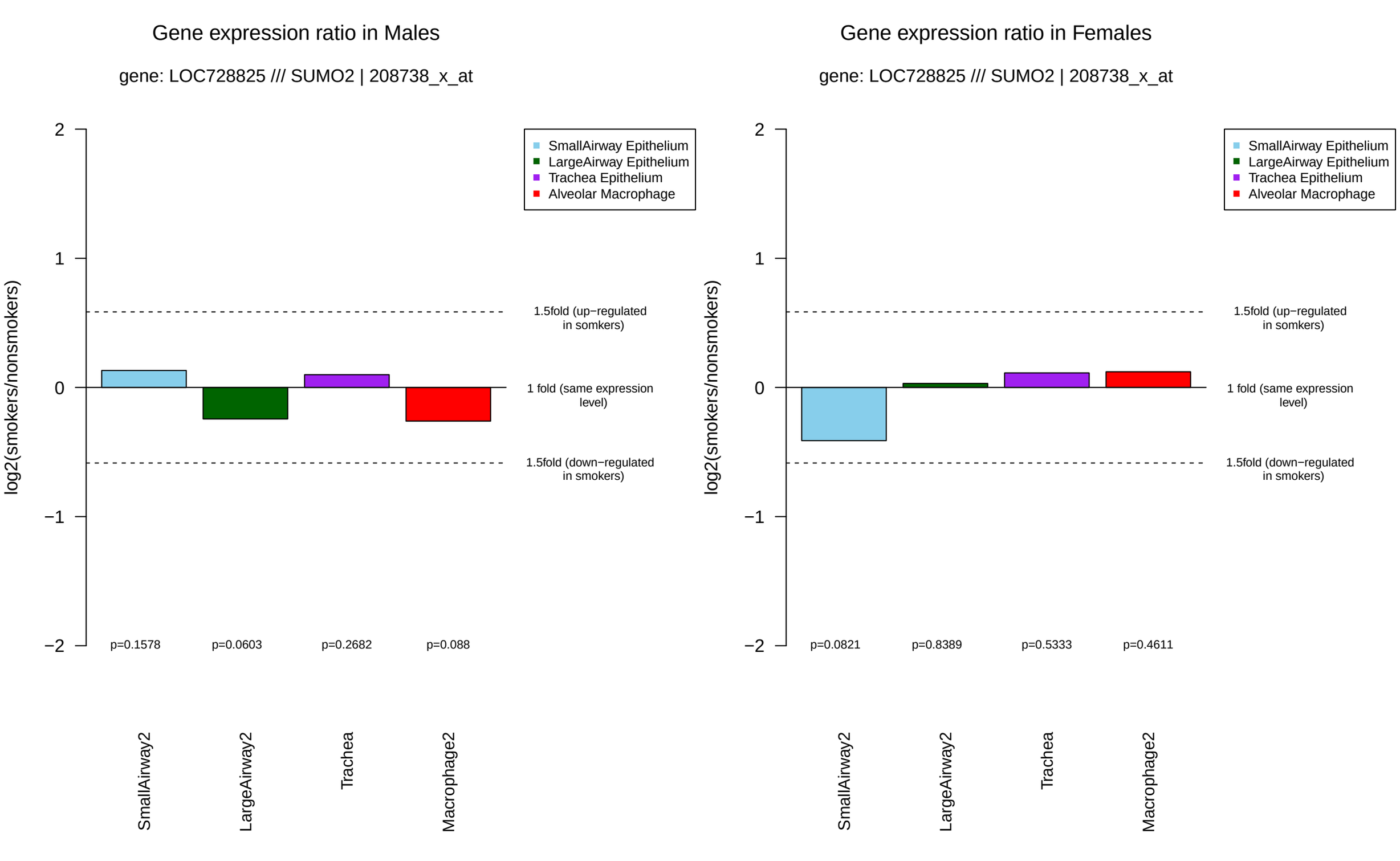
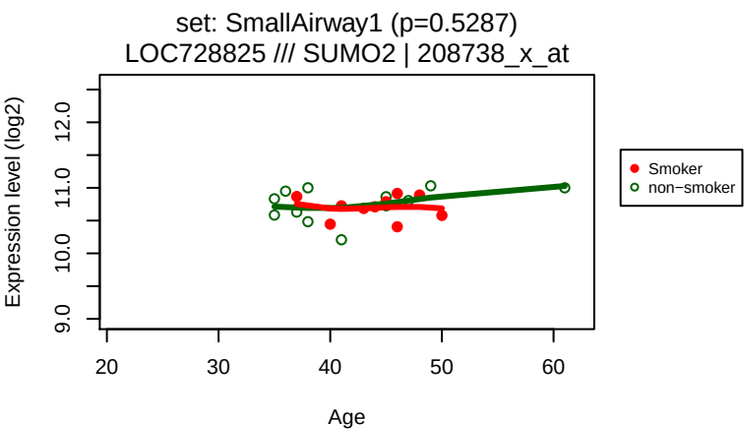
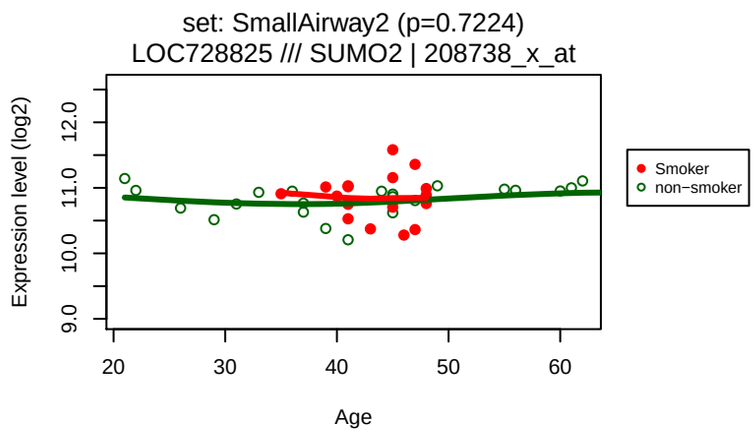
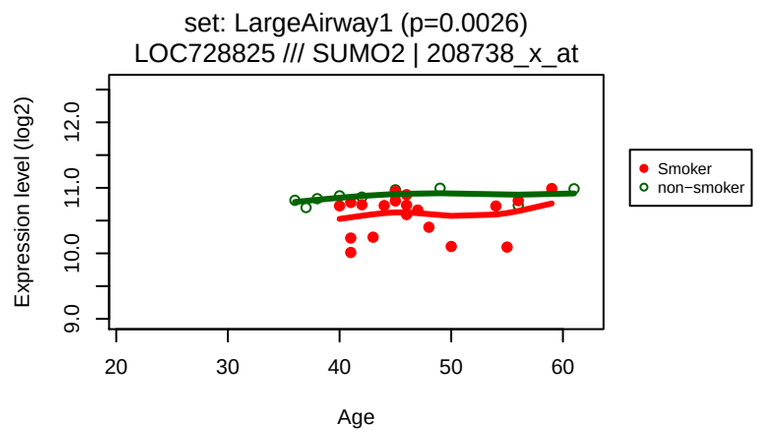

Gene expression of smokers and nonsmokers
gene: LOC728825 /// SUMO2 | 208738_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5287
p=0.7224
p=0.3012
p=0.0026
p=0.0961
p=0.2264
p=0.6717
p=0.2356
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

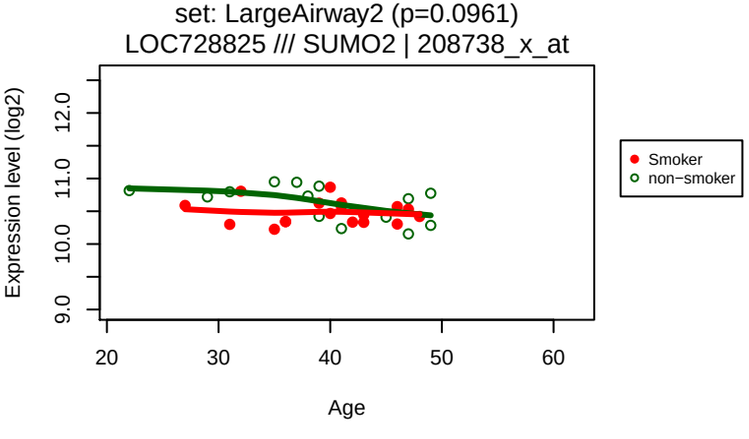

Gene expression ratio (smokers/nonsmokers)
gene: LOC728825 /// SUMO2 | 208738_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5287
p=0.7224
p=0.3012
p=0.0026
p=0.0961
p=0.2264
p=0.6717
p=0.2356
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

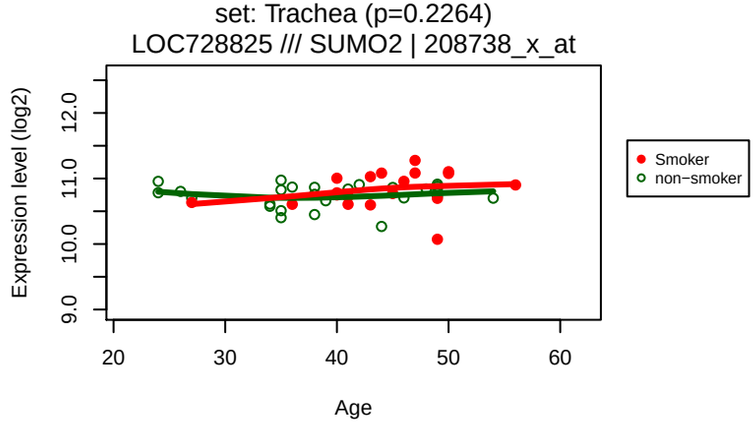

Gene expression ratio in Males
gene: LOC728825 /// SUMO2 | 208738_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1578
p=0.0603
p=0.2682
p=0.088
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC728825 /// SUMO2 | 208738_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0821
p=0.8389
p=0.5333
p=0.4611
SmallAirway2
LargeAirway2
Trachea
Macrophage2

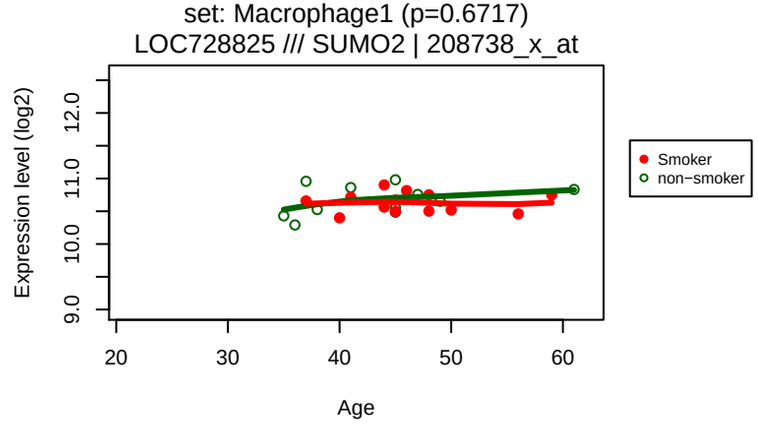

set: SmallAirway1 (p=0.5287)
LOC728825 /// SUMO2 | 208738_x_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker

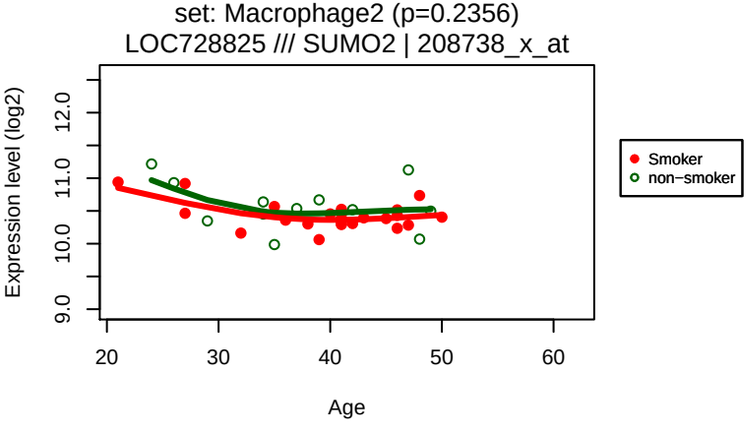

set: SmallAirway2 (p=0.7224)
LOC728825 /// SUMO2 | 208738_x_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker

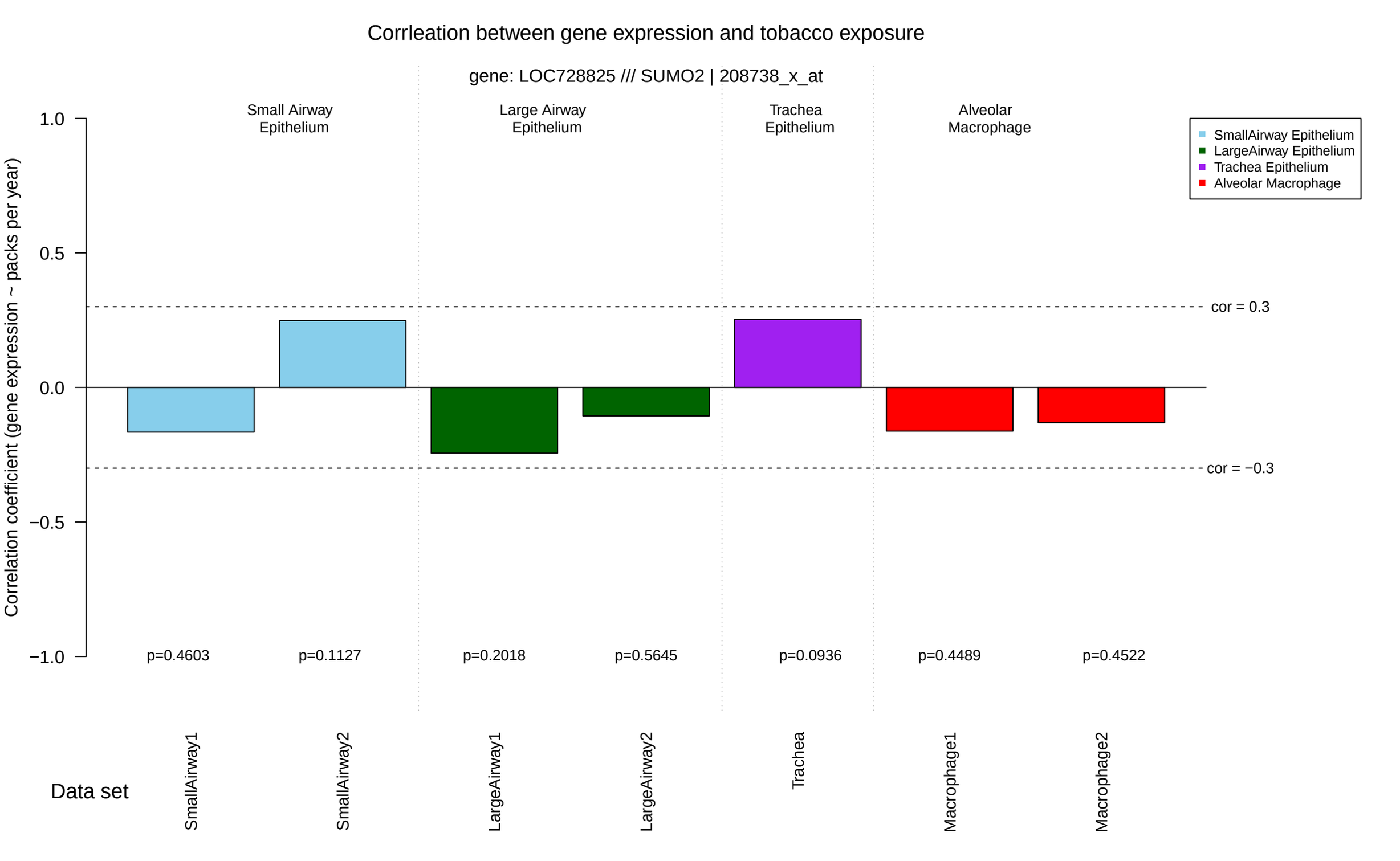

set: LargeAirway1 (p=0.0026)
LOC728825 /// SUMO2 | 208738_x_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0961)
LOC728825 /// SUMO2 | 208738_x_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2264)
LOC728825 /// SUMO2 | 208738_x_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6717)
LOC728825 /// SUMO2 | 208738_x_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2356)
LOC728825 /// SUMO2 | 208738_x_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC728825 /// SUMO2 | 208738_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4603
p=0.1127
p=0.2018
p=0.5645
p=0.0936
p=0.4489
p=0.4522
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2